Practical 4: Creating a Cube in SQL Server 2012

Aim

To create an OLAP Cube in SQL Server Analysis Services (SSAS) using Business Intelligence Development Studio (BIDS).

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What is a Cube? (Very Important)

A Cube (OLAP Cube) is a data structure used for fast data analysis.

It organizes data in multiple dimensions.

Example:

Sales Data Cube

Product	Year	Sales
Laptop	2023	500
Laptop	2024	700

Dimensions:

- **Product**
- **Time**

Measure:

- **Sales**

Cube helps analyze data very quickly.

Software Used

- **SQL Server 2012**
- **SQL Server Management Studio (SSMS)**
- **Business Intelligence Development Studio (BIDS)**

Step-by-Step Procedure

Step 1: Open BIDS

1. **Click Start Menu**
2. **Go to Microsoft SQL Server**
3. **Click SQL Server Business Intelligence Development Studio**

This opens the BIDS environment.

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Step 2: Create New Project

1. **Click File**
2. **Click New**
3. **Click Project**
4. **Select Analysis Services Project**
5. **Enter project name**
   Example:

SalesDataAnalysis

6. **Click OK**

Step 3: Create Data Source

1. **In Solution Explorer**
2. **Right click Data Source**
3. **Click New Data Source**
4. **Click Next**

Create Connection

1. **Enter SQL Server Name**
2. **Select Authentication**
3. **Select database**

Example:

Sales_DW

4. **Click Test Connection**
5. **Click OK**

Now the data source is created.

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Step 4: Create Data Source View

1. **Right click Data Source View**
2. **Click New Data Source View**
3. **Click Next**
4. **Select the Sales_DW database**
5. **Select FactProductSales table**
6. **Click Add Related Tables**
7. **Click Finish**

Now the Data Source View is created.

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Step 5: Create Cube

1. **Right click Cubes**
2. **Click New Cube**
3. **Click Next**

Select Tables

1. **Choose Use existing tables**
2. **Select FactProductSales**

Click Next

Select Measures

Measures are numeric values like:

- **Sales amount**
- **Quantity**
- **Total revenue**

Select the required measures.

Click Next

Select Dimensions

Dimensions are categories like:

- **Product**

- **Date**
- **Customer**

Select all dimensions.

Click Next

Name the Cube

Example:

SalesAnalyticalCube

Click Finish

Now the OLAP Cube is created.

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Step 6: Deploy the Cube

1. **Right click Project Name**
2. **Click Deploy**

Deployment sends the cube to SQL Server Analysis Services.

Step 7: Process the Cube

1. **Right click Project**
2. **Click Process**
3. **Click Run**

Now cube data is processed.

Step 8: Browse the Cube

1. **Right click Cube**
2. **Click Browse**

You can drag:

- **Product Name**
- **Date**
- **Sales**

to analyze the data.

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Result

An OLAP Cube is successfully created and deployed in SQL Server Analysis Services for data analysis.